Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	EV Industry Analytics & Intelligence Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static dataset: 1,200 synthetic EV records stored in CSV/JS files, no live data feed	Insights may not reflect the real-time EV market; limits accuracy and freshness of analysis	High
2	Web dashboard is client-side only (GitHub Pages) with no backend, database or login	Cannot support multi-user roles, private data or very large datasets	Medium
3	Tableau Public dashboard needs manual data extract refresh and has data size limits	Manual updates are time-consuming and not suitable for enterprise-scale data	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Static site on GitHub Pages + Tableau Public; read-only viewers	Host on cloud with CDN and load balancing; move to Tableau Cloud/Server for enterprise and multi-user access
2	Data Storage	CSV / JSON files with 1,200 records (20 columns)	Migrate to a cloud database (PostgreSQL / BigQuery) with automated ETL pipelines to handle millions of records
3	Performance	1,000 queries run in 7.6 ms with zero data errors	Add indexing, pre-aggregated summary tables, Tableau extracts and lazy-loading of charts to keep dashboards fast at scale
4	Security	Public, read-only data; no personal or sensitive information	Add authentication (SSO), role-based access, row-level security and HTTPS/encryption when private data is used

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Integrate live EV data sources (public APIs / datasets) with automated refresh	Q1 2027 (Jan - Mar)	Real-time insights and higher accuracy of market analysis
Phase 3	Move data to a cloud database and publish on Tableau Cloud/Server with role-based access	Q2 2027 (Apr - Jun)	Supports larger data volumes and many concurrent users securely
Phase 4	Add predictive analytics (sales and battery-tech forecasting) and a mobile-friendly view	Q3 - Q4 2027	Proactive decisions for OEMs, analysts and policymakers